

Heaps

1. (a) Draw the 11-entry hash table that results from using the hash function $h(i) = (3i + 5) \bmod 11$ to hash the keys 12, 44, 13, 53, 23, 71, 8, 39, 20, 16, and 5, assuming collisions are handled by *chaining* (or inserting into a list when there are collisions).
- (b) What is the result of the previous part, assuming collisions are handled by linear probing?
- (c) Show the result from part (a) assuming collisions are handled by quadratic probing (up to the point where the method fails).
- (d) Show the result from part (a) assuming collisions are handled by double hashing using a secondary hash function $h'(k) = 7 - (k \bmod 7)$.